ENCI335 Structural Analysis & Systems I
(II) Analysis of Indeterminate Structures
(3) Energy Methods for Structural Analysis

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Overview of Energy Methods

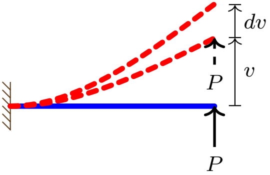
Overview of Energy Methods

- The energy concept is a fundamental principle in modern structural analysis, shifting the focus from local force equilibrium to global system behavior.
- It provides the theoretical foundation for the celebrated Finite Element Method, which is now the industry standard for simulating structural responses across all engineering disciplines.
- This approach replaces the tedious task of balancing forces at every joint with a single work-energy balance for the entire structure.
- By treating energy as a scalar quantity, it simplifies the analysis of complex, statically indeterminate structures that are often tedious to solve by hand.
- We will discuss the following topics in the coming lectures aiming to introduce you to this great concept.
  - Work Energy Principle (Conservation of Energy)
  - Betti's Law and Maxwell's Reciprocal Theorem
  - Castigliano's Theorems
  - Principle of Virtual Work
    - Principle of Virtual Displacement (Virtual Work)
    - Principle of Virtual Force (Complementary Virtual Work)
    - Unit-Load Method

Work-Energy Principle

External Work Energy W

In structural mechanics, work is done when a force causes an object to move in the direction of that force.



$dW = Pdv$

P increases slowly. (not a constant)

$W = \int Pdv$

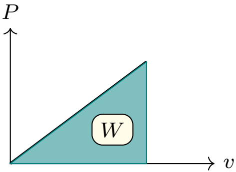
Similarly, for point moment: $W = \int Md\theta$

The area under the P-v response curve

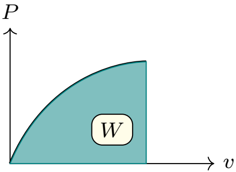
Linear Elastic Structure

Nonlinear Structure

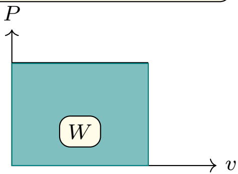
Constant P (v due to other actions)



$W = \frac{1}{2}Pv$



$W = \int_0^v Pdv$

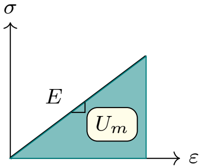


$W = Pv$

Internal Strain Energy U - Uniaxial Material

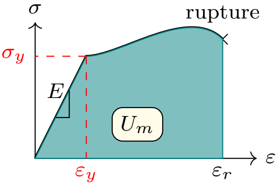
When a structure deforms under external loads, it gets strained/stressed and stores an internal strain energy, commonly denoted by U. For an elastic structure, this stored energy will be fully released once the load is removed, assuming no energy is lost due to friction, damping, etc.
To derive a mathematical expression for the strain energy stored in the structure, we start from the strain energy density U_m of a uniaxial material point (based on the material stress-strain curve), shown as follows:

Linear Elastic Material



$U_m = \frac{1}{2}\sigma\varepsilon$ $= \frac{\sigma^2}{2E}$

Nonlinear Material



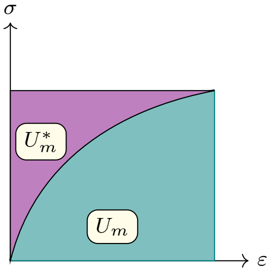
$U_m = \int_0^{\varepsilon} \sigma \, d\varepsilon$

Modulus of toughness:
$U_{mT} = \int_0^{\varepsilon_r} \sigma \, d\varepsilon$

Modulus of resilience:
$U_{mr} = \int_0^{\varepsilon_y} \sigma \, d\varepsilon$
$= \frac{\sigma_y \varepsilon_y}{2} = \frac{\sigma_y^2}{2E}$

Complementary Energy Density U_m^* for Uniaxial Materials

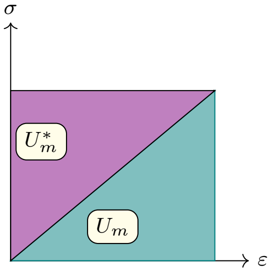
General Material



$U_m = \int_0^{\varepsilon} \sigma \, d\varepsilon$

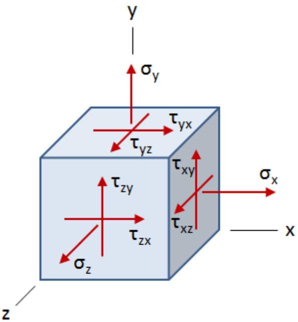
$U_m^* = \int_0^{\sigma} \varepsilon \, d\sigma$

Linear Material



$U_m^* = U_m = \frac{\sigma \varepsilon}{2}$

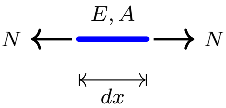
Strain Energy Density U_m - 3D Linear Elastic Material



$$U_m = \frac{1}{2}\left(\sigma_x\varepsilon_x + \sigma_y\varepsilon_y + \sigma_z\varepsilon_z + \tau_{xy}\gamma_{xy} + \tau_{xz}\gamma_{xz} + \tau_{yz}\gamma_{yz}\right)$$

Energy methods allow all stresses in the 3D space to be accounted for.

Strain Energy in a Prismatic Beam - Axial Component



$$\boxed{U_s = \int_A U_m \, dA = \int_A \frac{\sigma_x \varepsilon_x}{2} \, dA}$$

- For a linear elastic material, $\varepsilon_x = \sigma_x / E$.
- Based on the Saint-Venant's Principle: the normal stress σ_x is assumed uniform across the whole cross-section: $\sigma_x = \frac{N}{A}$.
- For a beam section, the strain energy is

$$U_s = \int_A \frac{N(x)^2}{2EA^2} \, dA = \frac{N(x)^2}{2EA^2} \int_A dA = \frac{N(x)^2}{2EA} \tag{1}$$

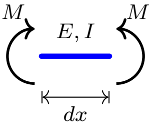
- For a beam of length L,

$$U_e = \int_0^L U_s \, dx = \int_0^L \frac{N(x)^2}{2EA} \, dx \tag{2}$$

- If N is constant, which is often the case,

$$U_e = \frac{N^2 L}{2EA} \tag{3}$$

Strain Energy in a Prismatic Beam - Moment Component



$$\boxed{U_s = \int_A U_m \, dA = \int_A \frac{\sigma_x \varepsilon_x}{2} \, dA}$$

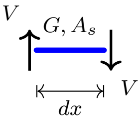
- For a linear elastic material, $\varepsilon_x = \sigma_x / E$.
- Based on the Euler-Bernoulli's assumption: plane section remains plane and normal to the beam axis after deformation. The normal stress σ_x distribution is linear across the whole cross-section: e.g. 2D: $\sigma_x(x) = \frac{-M(x)y}{I}$.
- For a beam section, the strain energy is

$$U_s = \int_A \frac{M(x)^2 y^2}{2EI^2} \, dA = \frac{M(x)^2}{2EI^2} \left(\int_A y^2 \, dA \right) = \frac{M(x)^2}{2EI} \tag{4}$$

- For a beam of length L,

$$U_e = \int_0^L U_s \, dx = \int_0^L \frac{M(x)^2}{2EI} \, dx \tag{5}$$

Strain Energy in a Prismatic Beam - Shear Component



$$\boxed{U_s = \int_A U_m \, dA = \int_A \frac{\tau_{xy}\gamma_{xy}}{2} \, dA}$$

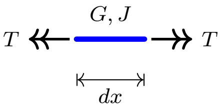
- For a linear elastic material, $\gamma_{xy} = \tau_{xy}/G$, where G is the shear modulus.
- Shear stress is often to be assumed weighted average across the section as: e.g. $\tau_{xy}(x) = \frac{-V}{A_s}$, where A_s is the shear area, not the actual cross-sectional area. Shear warping is ignored.
- For a beam section, the strain energy is

$$U_s = \int_A \frac{V(x)^2}{2GA_s^2} \, dA \approx \frac{V(x)^2}{2GA_s} \tag{6}$$

- For a beam of length L,

$$U_e = \int_0^L U_s \, dx = \int_0^L \frac{V(x)^2}{2GA_s} \, dx \tag{7}$$

Strain Energy in a Prismatic Beam - Torsion Component



$$\boxed{U_s = \int_A U_m \, dA = \int_A \frac{\tau\gamma}{2} \, dA}$$

- For a linear elastic material, $\gamma = \tau/G$.
- Based on the circular cross-section idea: the shear stress τ is linearly distributed from the centroid of the cross-section: $\tau = \frac{Tr}{J}$, where r and J are the radial distance and polar moment of inertia, respectively. Torsional warping is ignored.
- For a beam section, the strain energy is

$$U_s = \int_A \frac{T(x)^2 r^2}{2GJ^2} \, dA = \frac{T(x)^2}{2GJ^2} \int_A r^2 \, dA = \frac{T(x)^2}{2GJ} \tag{8}$$

- For a beam of length L,

$$U_e = \int_0^L U_s \, dx = \int_0^L \frac{T(x)^2}{2GJ} \, dx \tag{9}$$

- If T is constant, which is often the case,

$$U_e = \frac{T^2 L}{2GJ} \tag{10}$$

Application of Work Energy Principle

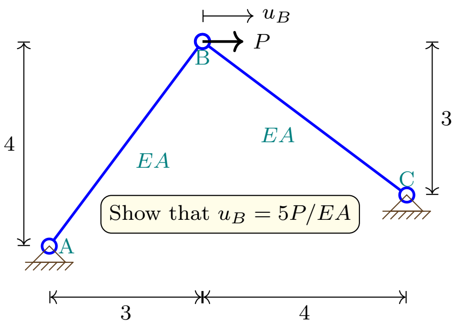
- Conservation of Energy

$$W = U \tag{11}$$

  This is the equation we use to obtain the displacement response of a structure.
- External Work W
  - Vertical point Load: $W = \frac{1}{2}Pv$
  - Horizontal point load: $W = \frac{1}{2}Pu$
  - Point Moment: $W = \frac{1}{2}M\theta$
- Internal Strain Energy U_e of a beam
  - Axial: $U_e = \int_0^L \frac{N(x)^2}{2EA}\,dx$
  - Moment: $U_e = \int_0^L \frac{M(x)^2}{2EI}\,dx$
  - Shear: $U_e = \int_0^L \frac{V(x)^2}{2GA_s}\,dx$
  - Torsion: $U_e = \int_0^L \frac{T(x)^2}{2GJ}\,dx$
- For a structure with multiple members and various energy components, we simply sum the contributions of all individual parts:

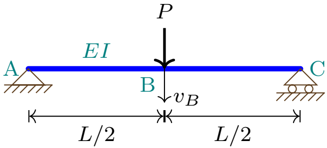
$$U = \sum U_e \tag{12}$$

Example 1 - Truss



Show that $u_B = 5P/EA$

Example 2 - Euler-Bernoulli Beam

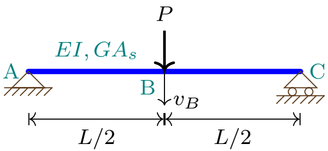


Show that $v_B = \frac{PL^3}{48EI}$

Assigning downward as the positive direction is acceptable, provided the convention is applied consistently.

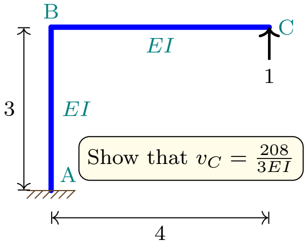
- In the energy method, work done is defined as positive when the force and its corresponding displacement act in the same direction, and negative otherwise.
- Because this relationship depends only on the relative direction of the vectors, the choice of sign convention is flexible, provided it remains consistent throughout the calculations.
- Therefore, while the standard convention defines upward as positive for load and deflection, downward may be assigned as the positive direction for convenience.
- Under any chosen convention, a negative calculated result simply indicates that the displacement occurs in the direction opposite to the applied force.
- This flexibility will be demonstrated in the following examples, where the sign convention is adjusted to suit the specific problem context.

Example 3 - Timoshenko Beam



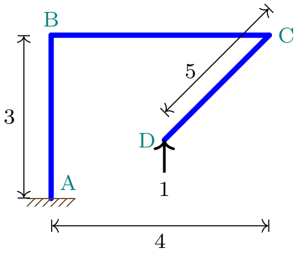
Is shear deformation significant?

Example 4 - Frame



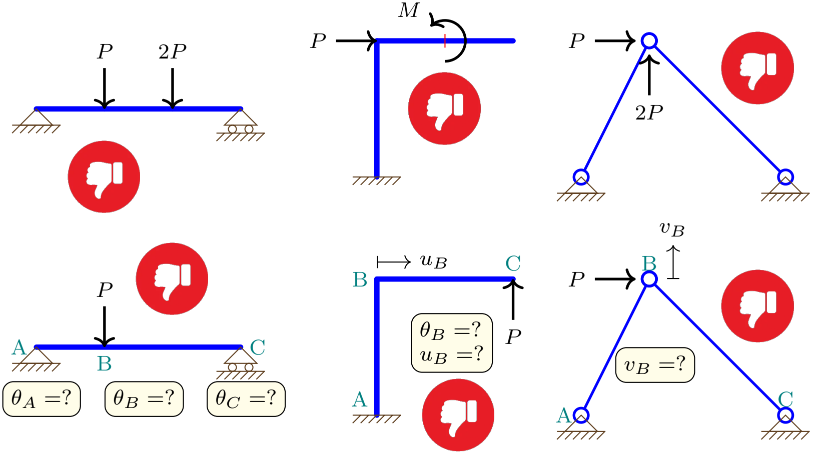
Show that $v_C = \frac{208}{3EI}$

Example 5 - 3D Frame



Limitations of Work-Energy Principle

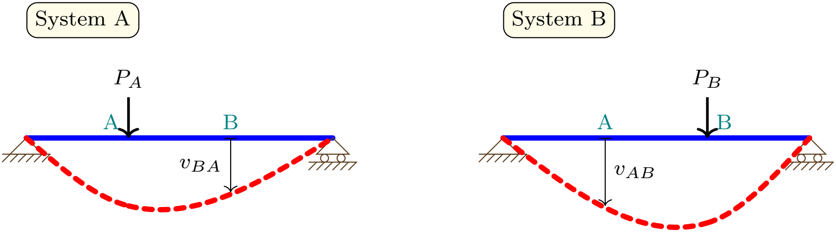
- Not solvable for more than one load is applied (1 equation, 1 unknown only).
- When a single load is applied, displacements or rotations at points or degrees of freedom (DOFs) not subjected to the load cannot be determined.



Betti's Law

Betti's Law (Proposed by Enrico Betti in 1872)

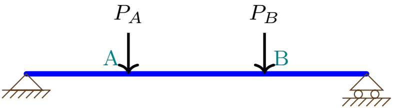
In a linear elastic structure, the work done by one system of forces and moments acting through the displacements and rotations caused by another system of forces and moments is equal to the work done by the second system acting through the displacements and rotations due to the first system.



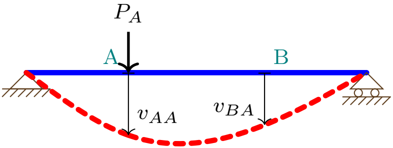
$$P_A v_{AB} = P_B v_{BA}$$

$$\sum_i P_{iA} v_{iB} = \sum_i P_{iB} v_{iA}$$

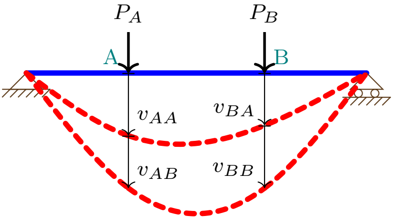
Proof of Betti's Law - 1/2



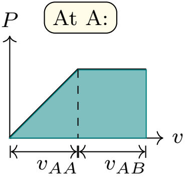
Sequence 1: applying P_A first

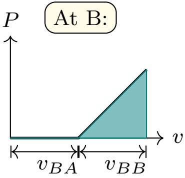


followed by applying P_B



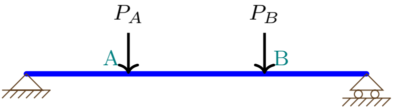
Use the idea that the external work done in an elastic structure is path independent!



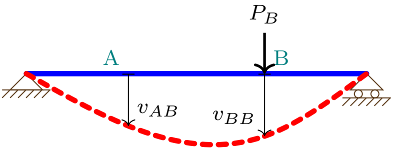


$$W_1 = \frac{1}{2}P_A v_{AA} + P_A v_{AB} + \frac{1}{2}P_B v_{BB}$$

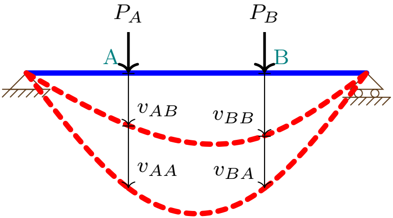
Proof of Betti's Law - 2/2



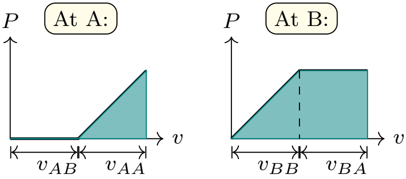
Sequence 2: applying P_B first



followed by applying P_A



Use the idea that the external work done in an elastic structure is path independent!



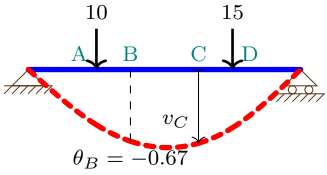
$$W_2 = \frac{1}{2}P_A v_{AA} + \frac{1}{2}P_B v_{BB} + P_B v_{BA}$$

$$W_2 = W_1 \implies P_A v_{AB} = P_B v_{BA}$$

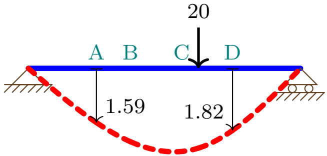
Q.E.D.
(quod erat demonstrandum)

Betti's Law - Examples

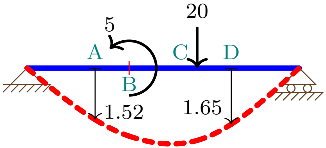
System 1



System 2



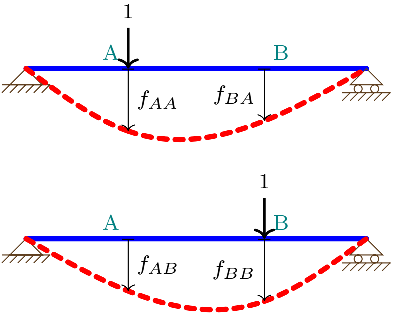
System 3



Maxwell's Reciprocal Theorem

Maxwell's Reciprocal Theorem

For a linear elastic structure, the displacement/rotation of B due to a unit load/moment at A is equal to the displacement/rotation of A when a unit load/moment is applied at B.



$f_{AB} = f_{BA}$

We use f because it is called flexibility.

Proof:

According to Betti's Law:

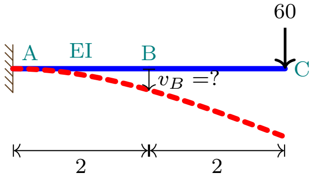
$P_A v_{AB} = P_B v_{BA}$

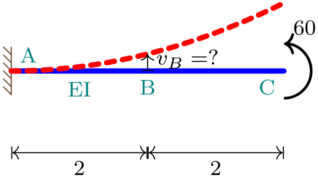
Divide both sides by $P_A P_B$:

$$\frac{v_{AB}}{P_B} = \frac{v_{BA}}{P_A}$$

$$f_{AB} = \frac{v_{AB}}{P_B},\ f_{BA} = \frac{v_{BA}}{P_A}$$

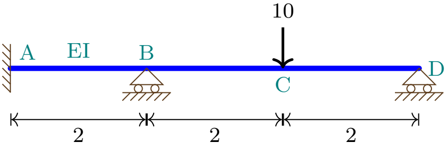
Q.E.D.


60
A
EI
B
C
v_B =?
2
2


60
A
v_B =?
EI
B
C
2
2

Maxwell's Reciprocal Theorem - Example 3

Confirm that $R_B = 10.45$ (upwards) and $R_D = 3.64$ (upwards).



Castigliano's Theorems

Castigliano's Theorems (by Alberto Castigliano in 1879)

- **First theorem** (applicable to both linear and nonlinear structures): The partial derivative of the strain energy (expressed as a function of displacements) with respect to a displacement component at a point is equal to the force corresponding to that displacement.

$$\frac{\partial U}{\partial v_i} = P_i \tag{13}$$

It can be used to derived equilibrium equations and leads to the displacement method, which is the most common method used in computer software for structural analysis (out-of-scope).

- **Second theorem** (applicable to linear structures only): The partial derivative of the strain energy (expressed as a function of forces) with respect to a force component applied at a point is equal to the displacement corresponding to that force.

$$\frac{\partial U}{\partial P_i} = v_i \tag{14}$$

It can used to derived compatibility conditions and leads to the force method, which will be discussed in this course.

Application of Castigliano's 2nd Theorems

Axial Components:

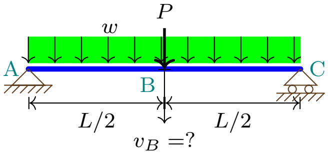
- $$U = \sum \frac{N^2 L}{2EA} \tag{15}$$
  $$v_i = \frac{\partial U}{\partial P_i} = \frac{\partial U}{\partial N}\frac{\partial N}{\partial P_i} = \sum \frac{NL}{EA} \cdot \frac{\partial N}{\partial P_i} \tag{16}$$
- $\frac{NL}{EA}$ is the member axial elongation, while $\frac{\partial N}{\partial P_i}$ is the axial force influence factor due to P_i (to be discussed later).

Bending Components:

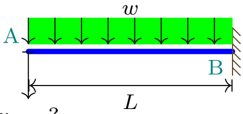
- $$U = \sum \int \frac{M^2}{2EI}\, dx \tag{17}$$
  $$v_i = \frac{\partial U}{\partial P_i} = \frac{\partial U}{\partial M}\frac{\partial M}{\partial P_i} = \sum \int \frac{M}{EI} \cdot \frac{\partial M}{\partial P_i}\, dx \tag{18}$$
- $\frac{M}{EI}$ is the section curvature, while $\frac{\partial M}{\partial P_i}$ is the bending moment influence factor due to P_i (to be discussed later).

For unloaded points, P_i is an added dummy load and set to zero at the final step.

Castigliano's 2$^{\text{nd}}$ Theorems - Example 1

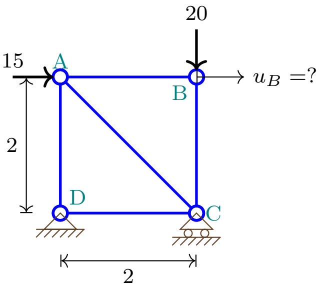


Castigliano's 2$^{\text{nd}}$ Theorems - Example 2



$v_A =?$
$\theta_A =?$

Castigliano's 2nd Theorems - Example 3



Principle of Virtual Work

Principle of Virtual Displacement (aka Principle of Virtual Work)

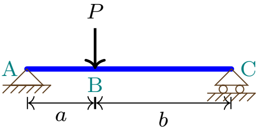
- If a structure is in equilibrium under a system of forces or moments and if it is subjected to any virtual but kinematically admissible displacements, the external virtual work done by the real external forces going through the virtual displacements is equal to the internal virtual work done by the real internal forces going through the virtual deformations.

$$\delta W = \delta U \text{ or } \sum P_i \cdot \delta v_i = \int_V \sigma \cdot \delta \varepsilon \, dV \tag{19}$$

where $\delta W = \sum P_i \cdot \delta v_i$ is the external virtual work, $\delta U = \int_V \sigma \cdot \delta \varepsilon \, dV$ is the internal virtual work, and δv and $\delta \varepsilon$ are the virtual displacements and strains satisfying the kinematic relation of the structure.

- This idea is widely used in the structural analysis software, but is beyond the scope of the course and will be discussed more in ENCI437.
- If the virtual displacement involves only rigid body motions, the external virtual work will be equal to zero, since there is no internal virtual work done by the internal forces.

PVD Examples



Principle of Virtual Force (aka Principle of Complementary Virtual Work)

- If a structure is in equilibrium under a system of forces or moments and if it is subjected to any set of virtual forces (both external and internal) satisfying equilibrium equations, the external complementary virtual work done by the virtual external forces going through the real displacements is equal to the internal complementary virtual work done by the virtual internal forces going through the real deformations.

$$\delta W = \delta U \text{ or } \sum \delta P_i \cdot v_i = \int_V \delta\sigma \cdot \varepsilon \, dV \tag{20}$$

where $\delta W = \sum \delta P_i \cdot v_i$ is the external complementary virtual work, $\delta U = \int_V \delta\sigma \cdot \varepsilon \, dV$ is the internal complementary virtual work, and δP and $\delta\sigma$ are the external and internal virtual forces satisfying equilibrium equations.

- This idea will be used in this course to solve for the response of statically indeterminate structures.
- If the virtual force involves only a unit load at one particularly degree-of-freedom (i.e. $\delta W = 1 \cdot v_i = v_i$), it helps us calculate the corresponding displacement. It also helps us solve for the unknown redundant forces of statically indeterminate structures. This method is called the unit-load method.

PVF Internal Virtual Work of Each Member

$$\delta U_e = \int_V \delta\sigma \cdot \varepsilon \, dV \tag{21}$$

- Axial component in a truss:

$$\delta U_e = \int_0^L n \cdot \varepsilon \, dx = n \cdot e, \text{ where } e = \frac{NL}{EA}, \tag{22}$$

  n is the axial force due to the virtual load δP, N is the axial force due to the real load P, and e is the real axial elongation in the member.

- Bending component in a beam:

$$\delta U_e = \int_0^L m \cdot \kappa \, dx, \text{ where } \kappa = \frac{M}{EI}, \tag{23}$$

  m is the bending moment due to the virtual load δP, M is the bending moment due to the real load P, and κ is the real curvature in the member.

The total virtual work for the full structure with multiple members is $\delta U = \sum \delta U_e$.

Unit-Load Method

Unit-Load Method for Trusses

- Determine displacement v_i for statically-determinate trusses

$$v_i = \sum n \cdot e = \sum \frac{n \cdot NL}{EA} \tag{24}$$

- For statically-indeterminate trusses
  - Identify redundant forces, say a support reaction R.
  - Enforce the compatibility condition. For example, if $SI = 1$, determine the support reaction R by solving the following

$$0 = \sum n \cdot e = \sum \frac{n \cdot (N + nR)L}{EA} \tag{25}$$

    where N is the axial force in each member due to the load case without the redundant force, and n is the axial force in each member due to a unit redundant force alone. The total axial force in each member is $N + nR$.
  - Rewriting the above equation gives the amount of the redundant force

$$R = -\frac{\sum \frac{nNL}{EA}}{\sum \frac{n^2 L}{EA}} \tag{26}$$

  - After R is obtained, we can apply the unit-load method on the structure with the redundant force removed to obtain the displacement v_i at all the degrees-of-freedom.

Unit-Load Method for Beams and Frames without Trusses

- Determine displacement v_i for statically-determinate beams and frames

$$v_i = \sum \int_0^L m \cdot \kappa \, dx = \sum \int_0^L \frac{m \cdot M}{EI} \, dx \tag{27}$$

- For statically-indeterminate beams and frames
  - Identify redundant forces, say a support reaction R.
  - Enforce the compatibility condition. For example, if $SI = 1$, determine the support reaction R by solving the following

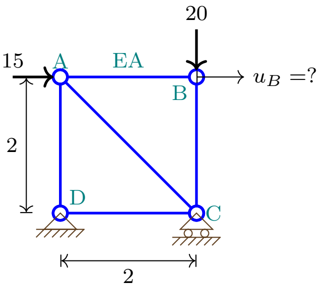
$$0 = \sum \int_0^L m \cdot \kappa \, dx = \sum \int_0^L \frac{m \cdot (M + mR)}{EI} \, dx \tag{28}$$

    where M is the bending moment in each member due to the load case without the redundant force, and m is the bending moment in each member due to a unit redundant force alone. The total bending moment in each member is $M + mR$.
  - Rewriting the above equation gives the amount of the redundant force

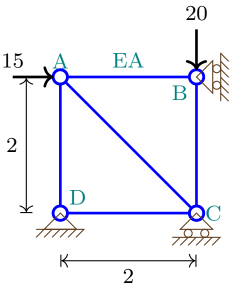
$$R = -\frac{\sum \int_0^L \frac{mM}{EI} \, dx}{\sum \int_0^L \frac{m^2}{EI} \, dx} \tag{29}$$

  - After R is obtained, we can apply the unit-load method on the structure with the redundant force removed to obtain the displacement v_i at all the degrees-of-freedom.

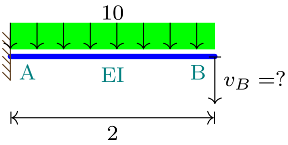
Unit-Load Method - Example 1 - Statically Determinate Truss



Unit-Load Method - Example 2 - Statically Indeterminate Truss

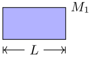
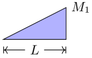
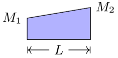
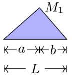
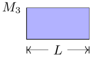
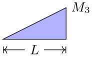
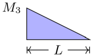
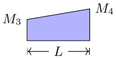
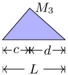
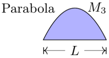
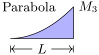


Unit-Load Method - Example 3 - Statically Determinate Beam

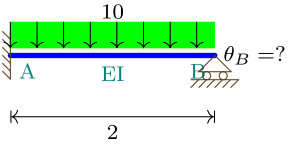


Integration Table (Leet et al.)

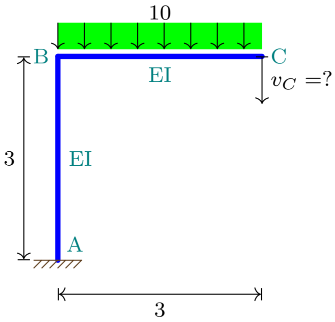
Table 1: Values of Product Integrals $\int_{x=0}^{x=L} M_Q M_P \, dx$ (adapted from Leet et al.)

M_Q / M_P				
	$M_1 M_3 L$	$\frac{1}{2} M_1 M_3 L$	$\frac{1}{2}(M_1 + M_2) M_3 L$	$\frac{1}{2} M_1 M_3 L$
	$\frac{1}{2} M_1 M_3 L$	$\frac{1}{3} M_1 M_3 L$	$\frac{1}{6}(M_1 + 2M_2) M_3 L$	$\frac{1}{6} M_1 M_3 (L + a)$
	$\frac{1}{2} M_1 M_3 L$	$\frac{1}{6} M_1 M_3 L$	$\frac{1}{6}(2M_1 + M_2) M_3 L$	$\frac{1}{6} M_1 M_3 (L + b)$
	$\frac{1}{2} M_1 (M_3 + M_4) L$	$\frac{1}{6} M_1 (M_3 + 2M_4) L$	$\frac{1}{6} M_1 (2M_3 + M_4) L$ $+ \frac{1}{6} M_2 (M_3 + 2M_4) L$	$\frac{1}{6} M_1 M_3 (L + b)$ $+ \frac{1}{6} M_1 M_4 (L + a)$
	$\frac{1}{2} M_1 M_3 L$	$\frac{1}{6} M_1 M_3 (L + c)$	$\frac{1}{6} M_1 M_3 (L + d)$ $+ \frac{1}{6} M_2 M_3 (L + c)$	for $c \le a$: $\left(\frac{1}{3} - \frac{(a-c)^2}{6ad}\right) M_1 M_3 L$
	$\frac{2}{3} M_1 M_3 L$	$\frac{1}{3} M_1 M_3 L$	$\frac{1}{3}(M_1 + M_2) M_3 L$	$\frac{1}{3} M_1 M_3 \left(L + \frac{ab}{L}\right)$
	$\frac{1}{3} M_1 M_3 L$	$\frac{1}{4} M_1 M_3 L$	$\frac{1}{12}(M_1 + 3M_2) M_3 L$	$\frac{1}{12} M_1 M_3 \left(3a + \frac{a^2}{L}\right)$

Unit-Load Method - Example 4 - Statically Indeterminate Beam



Unit-Load Method - Example 5 - Statically Determinate Frame



Unit-Load Method - Example 6 - Statically Indeterminate Frame